

A Universal Real-Time Contraction-Intensity Visualization System via sEMG Amplitude Normalization

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Abstract—Conventional XR interaction methods (controller buttons, hand tracking) fail to capture subtle muscular cues such as a user's intended grip force or contraction intensity. This study uses surface electromyography (sEMG) to classify hand contraction intensity in real time and visualizes it with color coding in Unity. Because EMG amplitude varies widely across subjects, a fixed threshold alone generalizes poorly; we combine a Logistic Regression model trained on multiple subjects to enable operation without per-user calibration. Leave-One-Subject-Out (LOSO) cross-validation was used for evaluation, revealing that random-split validation overestimates performance. Data were collected from four subjects using a MyoWare 2.0 sEMG sensor and an ESP32 microcontroller sampled at 100 Hz, and the model was implemented for real-time local inference in Unity. In LOSO validation, the Logistic Regression model achieved 86.9% accuracy, compared to 91.5% under random-split validation, confirming that subject-independent evaluation better reflects real-world generalization for EMG-based systems.

Research Keywords—sEMG, XR interaction, user variability, Logistic Regression, real-time visualization

1 INTRODUCTION

XR interaction mainly relies on controller buttons or hand-tracking-based gesture recognition. However, these methods cannot capture subtle muscular cues such as grip force, which limits fine manipulation of virtual objects. Surface electromyography (sEMG) can non-invasively measure muscle contraction intensity and therefore offers a potential input method for force-based XR interaction.

However, EMG signals show large amplitude variations across users, making fixed thresholds difficult to generalize. User-specific calibration can address this issue, but it reduces practical usability. Therefore, the key challenge is to recognize contraction intensity in real time while minimizing the need for prior user calibration.

This study implements a real-time XR interaction system that classifies hand contraction intensity from single-channel sEMG and visualizes it through color changes in Unity. To improve generalizability, we combine a threshold-based path with a lightweight Logistic Regression model trained on pooled multi-subject data. We also apply Leave-One-Subject-Out (LOSO) cross-validation to evaluate realistic performance for unseen users and to show that random-split validation can overestimate biosignal classification performance.

2 RELATED WORK OR LITERATURE REVIEW

Wearable EMG has been widely studied for hand gesture recognition. For example, previous studies have used wrist-worn EMG and IMU sensors or armband-type EMG sensors to classify hand gestures in real time. These works are related to this study in their use of wearable biosignals for interaction.

However, most prior research focuses on discrete gesture classification rather than contraction intensity as a continuous

or stepwise input. Many approaches also require user-specific training or calibration, which limits immediate use by new users. In contrast, this study focuses on hand contraction intensity, uses pooled multi-subject data to reduce calibration requirements, visualizes the result in real time in an XR environment, and evaluates generalization using subject-independent LOSO validation.

3 CONCEPT, MODEL AND METHODOLOGY

3.1 Concept and Model

The proposed system uses a dual-path structure combining a threshold-based path and a machine-learning-based path. Path A applies fixed thresholds derived from four subjects' training data and rapidly classifies sEMG signals into four levels: REST, LIGHT, STRONG, and GRIP. This path supports fast timing decisions. Path B uses a pooled probability model to reduce the effect of individual amplitude differences. By combining the two paths, the system aims to achieve both real-time responsiveness and improved generalizability.

3.2 Methodology

Data were sampled at 100 Hz using a MyoWare 2.0 EMG sensor and an ESP32 with a 12-bit ADC. Binary rest/grip data were collected from four subjects. The raw mean grip values varied notably across users; for example, s2 showed 2599 while s3 showed 1108, indicating an approximately 2.3-fold amplitude difference.

Five time-domain features were extracted from a 200 ms sliding window with 50% overlap: MAV, RMS, STD, WL, and ZC. The features were standardized using StandardScaler and classified with Logistic Regression. The model is lightweight, consisting

of five coefficients and one intercept, making it suitable for real-time local implementation in Unity using C#.

For validation, random train/test split and Leave-One-Subject-Out (LOSO) cross-validation were compared. LOSO excludes one subject from training and tests the model on that unseen subject, making it more appropriate for evaluating generalization in EMG data.

4 EXPERIMENTES, SIMULATIONS, AND/OR ANALYSES

4.1 Experiments and System Implementation

First, amplitude variation across subjects was examined in the collected data. In the grip condition, the raw mean value was 2599 for s2 and 1108 for s3, showing a maximum difference of approximately 2.3 times.

Table 1. Mean raw rest/grip values per subject

Subject	rest	grip
s1	18	1321
s2	38	2599
s3	0	1108
s4	10	1419

In LOSO validation, RandomForest achieved a generalization accuracy of 89.2%, while Logistic Regression achieved 86.9%. Subject s3 showed relatively low performance, ranging from 67% to 75%, which is presumed to be caused by electrode-contact variability. Although RandomForest performed slightly better, Logistic Regression was selected as the final model because of its lightweight linear structure with five coefficients and its ease of real-time implementation in Unity using C#.

The final coefficients were $\text{coef} = [1.2128, 2.0272, 0.0914, -0.4476, 0.9471]$ with an intercept of -0.0252 . These were implemented for local real-time inference without separate server communication, allowing new users to use the system immediately with their own muscle signals without additional setup.

The implemented model is connected in real time to Unity's XR interaction environment. EMG signals received from the ESP32 are classified into four levels through Path A, and the result is immediately visualized as a color change on a box object in the scene. Users can interact with the XR environment using only muscle signals, without operating a separate controller. In a preliminary validation test with one participant, the ML model's probability reached approximately $p \approx 0.98$ during maximum contraction, matching the threshold-based GRIP decision.

4.2 Analysis

The random split produced an accuracy of 91.5%, whereas LOSO validation resulted in 86.9%. This difference is interpreted as an overestimation caused by temporal leakage between adjacent windows. The result quantitatively shows that, for biosignals such as EMG, where inter-subject variation is large, random-split validation can overestimate actual performance. Therefore, subject-independent validation should be used as the standard for reporting performance in such signals.

Meanwhile, in the LIGHT-to-STRONG range, the ML model remained in the REST state and switched to GRIP only around p

≈ 0.8 , reacting more conservatively than the decision threshold of 0.5. This is presumed to be because the training data consisted only of two classes, rest and grip, resulting in low sensitivity to intermediate intensity levels. This observation supports the design decision to use Path A as the primary decision path and Path B as an auxiliary verification path.

5 CONCLUSIONS

This study implemented an XR interaction system that classifies hand contraction intensity in real time using single-channel sEMG and visualizes it through color changes in Unity. To overcome amplitude variation across subjects, we proposed a lightweight pooled model and confirmed its generalization performance using LOSO validation. By combining a threshold-based path and an ML-based path, the system achieved both precise timing and universal discrimination.

6 ACKNOWLEDGMENT

This research was supported by the MSIT (Ministry of Science and ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP (Institute of Information & Communications Technology Planning & Evaluation) in 2026 (2023-0-00089).

7 REFERENCES

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